

**Ismail Aissaoui**

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**Prof. Emmanuel Dubois & Prof. Marcos Serrano** (Univ Toulouse)

**Assoc. Prof. Cédric Fleury** (IMT Atlantique)

**Dr. Louis-Pierre Bergé** (CeaTech)

**Subject: Application for PhD Position – Supporting transitions for interacting in Digital Twins (PC5)**

Dear Prof. Dubois, Prof. Serrano, Prof. Fleury, and Dr. Bergé,

I am applying for the PC5 PhD position on "Supporting transitions for interacting in Digital Twins" with great enthusiasm. With a solid engineering foundation in human-computer interaction (HCI), multi-modal system design, and 3D web visualizations, I am highly motivated to tackle the challenge of seamlessly transitioning user interactions across varying levels of immersion in digital twin environments.

In my engineering work, particularly when architecting the "Live Control Room," I recognized that users require different interfaces depending on their operational context. While a technician on the factory floor might benefit from an augmented reality (AR) overlay to inspect a turbine, a remote engineer needs a comprehensive, non-immersive desktop dashboard. To address this, I developed a unified state-management architecture utilizing Next.js and Three.js that synchronized data representations across different visual contexts. This experience positions me perfectly to formalize interaction transitions in your project. I am eager to research novel ways to maintain cognitive continuity and contextual awareness when a user shifts from a 2D screen to an immersive mixed reality environment, ensuring the physical referent remains clearly linked.

I am particularly drawn to this position because it addresses a critical gap in the usability of modern digital twins. I look forward to validating these transitional frameworks across the diverse industrial use cases provided by the EDT program and contributing to the Artemis platform. Please find attached my resume and academic transcripts. I welcome the opportunity to discuss my application further.

Sincerely,

**Ismail Aissaoui**